INCH-POUND

MIL-C-60327A (AR)
AMENDMENT 2
20 APRIL 1990
SUPERSEDING
AMENDMENT 1
10 July 75

MILITARY SPECIFICATION

CHARGE ASSEMBLY, EXPELLING,
FOR CARTRIDGE, 105MM, ILLUMINATING, M314A2E1

This Amendment forms a part of Military Specification MIL-C-60327A (MU) dated 16 September 1966, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, change:

From: "CHARGE ASSEMBLY, EXPELLING, FOR CARTRIDGE, 105MM, ILLUMINATING, M314A2E1"

To: "CHARGE, EXPELLING, FOR 105MM, BASE EJECTION, CARTRIDGES"

1.1, change:

From: "This specification covers one type of expelling charge assembly for use in 105MM, Illuminating, M314A2E1 Cartridge."

To: "This specification covers one type of expelling charge assembly for use in 105MM, Illuminating, M314A2E1 Tactical, CS, XM629 and Smoke, M84E1 Cartridges."

* 2.1, add the following new specification to Specifications, Military:

"MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for"

2.1, add the following new specification to Specifications, Military:

"MIL-P-116 - Preservation, Methods of"

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2.1, delete the following from Standards, Military:

"MIL-STD-129 - Marking for Shipment and Storage"

2.1, add to Drawings, Army the following new drawing:

"8796522 - Marking Diagram and sealing for wood packing boxes"

PAGE 2

2.2, change the publication in Other publications:

From: "Code of Federal Regulations

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Rules and Regulations are now a part of the Code of Federal Regulations (1949 Edition and Revisions),available from the Superintedent Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-90" (latest revision).)"

To: "Code of Federal Regulations

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify CFR - 0-190 (latest revisions)."

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3.3, change the black powder moisture content requirement from "0.3 percent" to "0.70 percent"

PAGE 4

4.1.2, delete in its entirety.

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* 4.3, change title:

From: "Inspection provisions"

To: "Quality conformance inspection."

4.3.1: Change paragraph from:

"Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety,performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall not be less than the smallest weekly estimate of quantities contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector."

To: "Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of expelling charges shall contain:

a. Cups of one lot interfix number from not more than one manufacturer.

b. Covers of one lot interfix number from not more than one manufacturer.

c. Black powder from not more than one manufacturer."

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* 4.3.2: Add "MIL-A-48078 applies." to the beginning of the existing paragraph.

* 4.3.2: Add the following new sentence to the end of the existing paragraph:

"Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with this contract."

* 4.3.2.1: Change the Major category inspection level from "None Defined" to " Paragraph 4.5.1

4.3.2.1: Add the following new Major category:

"101. Thickness at base.....Gage 01006"

* 4.3.2.1: Change inspection level of Minor defects from "AQL 2.5 percent" to "Paragraph 4.5.1"

4.3.2.1: Change category 201 from "201 Thickness...Gage... 01001" to "201 Thickness of sidewall.....Gage...01001"

* 4.3.2.2: Change the "Major" heading from "Major AQL 0.40 percent" to "Major Paragraph 4.5.1"

* 4.3.2.2: Change the "Minor" heading from "Minor AQL 2.5 percent" to "Minor Paragraph 4.5.1"

4.3.2.2: Change category 202

From: "202 Outside diameter...Gage 02003"

To: "202 Outside diameter at base...Gage 02003"

4.3.2.2: Add new Minor Defect 208:

"208. Outside diameter at top.....Gage 02009"

* 4.3.2.4, change the Major section

From: "Major AQL 0.40 percent
101 Assembly incompletely sealed..Visual/Manual 04001"

To: "Major 100% Inspection, Paragraph 4.4.2
101 Assembly incompletely sealed..Gage 04001"

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* 4.3.2.4, change the "Minor" section

From: "Minor AQL 1.5 percent"

To: "Minor AQL 1.0 percent"

4.3.2.4, delete Minor Defect 201 in its entirety.

4.3.2.4, delete Minor Defect 202 in its entirety.

4.4.1, delete the last sentence in its entirety.

* 4.4.2, change paragraph, "Heat seal test of expelling charge assembly"

From: "The procedure for performing this test shall be as indicated in PA drawing 9221798. However, the leakagerate shall be no greater than .4 atmospheric cc/sec in lieu of .01 atmospheric cc/sec as cited on the drawing."

To: "Each assembly subjected to this test shall be placed in a transparent pressure vessel which contains a sufficient amount of water so that the assembly is completely immersed to a minimum depth of one inch. Units tested shall be oriented and arranged so that expelling air is visible to the inspector and able to reach the surface without obstruction. Once immersed, the pressure of the vessel shal be reduced from atmospheric by three inches of mercury. The pressure reduction shall be held for a minimum of twenty seconds. An indication of air bubbles emitting from the expelling charge assembly at any time after immersion shall be cause for rejection with the unit classified as nonconforming and removed from the lot. The entire test duration time shall not be longer than three minutes from the time of immerision."

* 4.5.1, add the following new paragraph:

"4.5.1 Sampling and inspection. In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time has been established (see Note 1), three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a process control, the production sample shall be inspected for the defects listed in paragraph 4.3 during continuous production. A new sample group will be taken when the first of the following events occurs:

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A. Five thousand parts have been produced

B. Five business days of production

C. Any change in process

D. Any change in material or material lot

Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. Further, that portion of production since the last control check shall be inspected for each separate type of defect according to MIL-STD-105, using an AQL of 0.40percent for each major defect and an AQL of 2.5 percent for each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

Note 1: In establishing a cure time, dimensionally check five (5) parts from each cavity at periodic intervals (e.g. every 5, 10, etc. min.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established."

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5.1, change title from "Packing" to "Preservation and Packaging"

5.1.1, change section

From: "Level C. Expelling charge assemblies shall be packed in accordance with the Code of Federal Regulations "49 CFR 71-90" and to afford protection against damage duringshipment from the supply source to the first receivingactivity for immediate use. Containers shall conform to the carrier rules and regulations applicable to the mode of transportation."

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To: "Level C. The expelling charges shall be packaged in accordance with the Code of Federal Regulation "49 CFR 0-190" and in a manner which will afford protection against damage during shipment from the supply source to the first receiving activity for immediate use."

5.2, change

From: "Marking. In addition to any marking required by the contract or order, marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations "41 CFR 71- 90".

To: "Packing"

5.2.1, add the following new subparagraph, Level C:

"5.2.1 Level C. The expelling charges shall be packed in accordance with the Code of Federal Regulation "49 CFR 0-190" and in a manner which will afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers and Pack shall conform to the carrier rulesand regulations applicable to the mode oftransportation."

5.3, add new subparagraph, "Marking".

5.3.1, add the following new subparagraph:

"In addition to any marking required by the contract or order, marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations "41 CFR 0-190" and dwg. 8796522. The DOT marking shall be "BLACK POWDER"."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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